Traceable Hybrid Auditing of Clinical LLM Outputs: Architecture and Pilot Calibration Combining Span Verification, Protocol Rules, and Graph Kernels

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Abstract

Background: Evaluating Large Language Models (LLMs) in open-ended clinical workflows such as acute chest pain triage requires auditing procedural adherence (omissions, temporal ordering, and contraindications) rather than static multiple-choice accuracy. Unconstrained *LLM-as-a-Judge* paradigms risk unverified event extraction and often penalize appropriate diagnostic uncertainty.
Objective: We present the software architecture of a **Traceable Neuro-Symbolic Auditing Framework** and demonstrate its mechanism calibration on a pilot development set ($N = 10$ traces, plus 5 targeted linguistic unit tests).
Methods: The pipeline decouples auditing into three layers: (1) a `TraceExtractor` enforcing deterministic verbatim span verification and epistemic assertion calibration (`PROPOSED`, `REPORTED_PERFORMED`, `NEGATED`, `SUSPECTED`, `DEFINITIVE`), (2) a `ClinicalRuleEvaluator` checking guideline-derived temporal ordering and contraindications, and (3) a graph-based `Reasoning Kernel` detecting unsupported definitive claims (`Pattern RP-001`).
Pilot Calibration Results: On our 10-case development suite (5 synthetic protocol-violation/control traces and 5 real frontier LLM responses), introducing assertion-status calibration eliminated 3 false-positive flags on negated actions and suspected differentials (shifting development specificity from 4/7 to 7/7 while preserving 3/3 violation detection). All 5 targeted linguistic stress unit tests passed as designed.

1 Introduction

While Large Language Models (LLMs) perform strongly on static medical licensing examinations [4], open-ended emergency triage of **Acute Chest Pain** [1] requires strict procedural adherence. An LLM may list a correct differential diagnosis while proposing unsafe management steps, such as discharging a patient after symptomatic relief from a gastrointestinal cocktail prior to reviewing serial troponins (premature closure), or administering empiric heparin in suspected aortic dissection.

Current *LLM-as-a-Judge* evaluation methods [5] face two structural limitations in procedural clinical auditing:

1. **Unverified Event Extraction:** A neural judge may infer or hallucinate that a candidate response ordered an investigation when no such statement exists in the text.

2. **Epistemic Conflation:** Building upon classical clinical negation and context detection principles established in `NegEx` [2] and `ConText` [3], an effective auditor must distinguish negated warnings (*"Do not give heparin"*) and suspected differentials from definitive clinical commitments.

2 System Architecture

Our framework separates linguistic extraction from deterministic symbolic verification across three tiers:

2.1 Tier 1: `TraceExtractor` (Span Firewall & Epistemic Tagging)

Given raw text T, candidate events $E = \{e_1, \ldots, e_n\}$ must satisfy verbatim substring verification under whitespace/case normalization $\mathcal{N}(s_i) \subseteq \mathcal{N}(T)$, rejecting ungrounded extractions (`SPAN_NOT_IN_SOURCE`). Lexical synonyms are mapped to canonical protocol codes, and events are filtered by assertion status $\alpha(e_i)$ so that `NEGATED` warnings are never penalized as proposed actions, and only `DEFINITIVE` inferences require structural evidentiary proof.

2.2 Tier 2 & Tier 3: `ClinicalRuleEvaluator` & `Reasoning Kernel`

The rule engine evaluates mandatory omissions (`CRITICAL_OMISSION`), prerequisite temporal ordering ($a_{\text{pre}} \prec a_{\text{target}}$) for premature closure, and contraindications (`must_avoid`) mapped to 2021 AHA/ACC chest pain guidelines [1]. Concurrently, the `Reasoning Kernel` constructs a Directed Acyclic Graph $G = (V, E_{\text{graph}})$ and triggers `RP-001` whenever a `DEFINITIVE` inference node has support in-degree $\deg^-_{\text{SUPPORTS}}(v) = 0$.

3 Development Calibration & Unit Verification

Table 1 reports the development mechanism calibration across $N = 10$ traces alongside 5 targeted linguistic unit tests (`run_edge_cases.py`) testing quoted errors, conditional discharge, fabricated spans, lexical synonyms, and supported inferences.

Table 1: Development Mechanism Calibration on $N = 10$ Traces (3 Synthetic Positives, 7 Negatives)

Auditor Configuration	TP	FP	TN	FN	Dev Recall (3/3)	Dev Specificity (TN/7)	Dev F_1
1. Uncalibrated Baseline	3	3	4	0	100.0% (95% CI: 29.2–100%)	57.1% (95% CI: 18.4–90.1%)	0.667
2. Calibrated Hybrid Pipeline	**3**	**0**	**7**	**0**	**100.0% (95% CI: 29.2–100%)**	**100.0% (95% CI: 59.0–100%)**	**1.000**

4 Methodological Limitations

This document describes our v0.1 software architecture and development calibration. Because the rules and test cases were co-developed on a small sample ($N = 10$, with 3 synthetic positive violations), these figures represent functional verification (unit testing) rather than generalizable empirical performance. Furthermore, while substring verification confirms verbatim quote existence, assertion status assignment remains dependent on upstream parser classification accuracy. A formal comparative evaluation against standalone *LLM-as-a-Judge* baselines on held-out multi-model responses is conducted in our expanded study.

References

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